

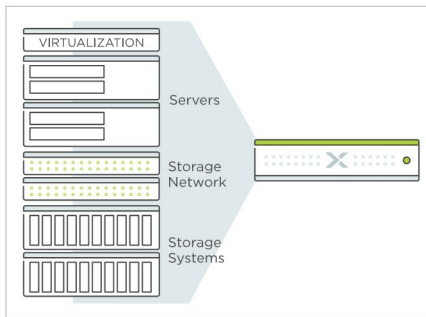


Figure 2: Hyper-convergence combines computing, storage, networking and virtualisation into one easy-to-handle software-defined system (Courtesy: Nutanix)

The bring-your-own-hardware kind

According to a recent Gartner report, software-only 'bring-your-own-hardware' hyper-converged systems have become significant and are increasingly competing with hyper-converged hardware appliances.

Gartner research director Jon MacArthur explained in a Web report that, "Most of the core technology is starting to shift to software. If you buy a vSAN ReadyNode from Lenovo, or Cisco, or HPE, or your pick of server platform, they are all pretty much the same. Customers are evaluating vSAN and ReadyNode, not so much the hardware. Dell EMC VxRAIL is a popular deployment model for vSAN, but so is putting VMware on your own choice of hyper-converged hardware."

Special needs like those of the military and the sophistication level of users are some of the factors that influence the decision in favour of implementing HCI as a software stack rather than a prepackaged solution.

Major players

Gartner predicts that the market for hyper-converged integrated systems (HCIS) will reach nearly US\$ 5 billion

(24 per cent of the overall market for integrated systems) by 2019 as the technology becomes mainstream. So, apart from specialists like Nutanix and Pivotal3, we also see industry majors like Cisco, Dell and HPE entering the space with their own or acquired technologies.

Nutanix, a pioneer in this space, offers a range of hardware systems with different options for the number of nodes, memory/storage capacity, processors, etc. These can run VMware vSphere or Nutanix's own hypervisor system called Acropolis. The company excels at supporting large clusters of HCI deployments, scaling up to 100 node clusters, which are easy to use and manage.

Nutanix tempts customers with the idea of a full-stack infrastructure and platform services with the promise of 'One OS, One Click'. Prism is a comprehensive management solution, while Acropolis is a turnkey platform that combines server, storage, virtualisation and networking resources. Calm is for application automation and life cycle management in Nutanix and public clouds, while Xpress is designed for smaller IT organisations. Xi Cloud Services help extend your data centre to an integrated public cloud environment.

Nutanix also has a community edition that lets you evaluate the Nutanix Enterprise Cloud Platform at no cost!

Companies like Cisco and HPE are strengthening their foothold by acquiring startups in the space. Cisco bought Springpath, whose technology powers Hyperflex—Cisco's fully-integrated hyper-convergence infrastructure system. HPE acquired Simplivity, one of the major competitors of Nutanix. HPE Simplivity products come with a strong, custom-designed platform called OmniStack, which includes a host of features like multi-site data management, global deduplication, backup, snapshots, clones, multi-site data replication, disaster recovery and WAN optimisation.

Pivotal3 also finds a place amongst the most popular HCI players. Last year, the company launched Acuity, which it claims to be the first priority-aware, performance-architected HCI solution with policy-based management. Acuity's advanced quality of service (QoS) makes it possible to simply and predictably run multiple, mixed applications on a single platform.

Dell EMC offers the VxRail and XC series of HCI solutions based on its PowerEdge servers powered by Intel processors. Last year, it released hyper-converged appliances that include Intel's 14-nanometre Xeon SP processors, along with VMware's vSAN and EMC's ScaleIO. Dell EMC offers NVIDIA GPU accelerators in both VxRail and XC solutions.

Apart from the regular HCI infrastructure, some startups are also coming up with innovative solutions, which some trend-watchers describe as HCI 2.0! NetApp, for example, has an option for those who want to keep their servers and storage separate—either to share the storage with non-HCI systems or to offload certain tasks to dedicated servers. NetApp HCI uses SolidFire technology to deliver clusters with dedicated storage and compute nodes.

Companies like Pivotal3, Atlantis Computing and Maxta also offer pure software HCI solutions.